

EX.NO:4 CREATE AN ABSTRACT CLASS – SHAPE CLASS

PROGRAM:

```
import java.util.*;
abstract class Shape {
    public int x, y;
    public abstract void printArea();
}
class Rectangle1 extends Shape {
    public void printArea() {
        float area;
        area = x * y;
        System.out.println("Area of Rectangle is " + area);
    }
}
class Triangle extends Shape {
    public void printArea() {
        float area;
        area = (x * y) / 2.0f;
        System.out.println("Area of Triangle is " + area);
    }
}
class Circle extends Shape {
    public void printArea() {
        float area;
        area = (22.0f * x * x) / 7.0f;
        System.out.println("Area of Circle is " + area);
    }
}
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Length and Breadth: ");
        int x1 = sc.nextInt();
        int y1 = sc.nextInt();
        Rectangle1 r = new Rectangle1();
        r.x = x1;
        r.y = y1;
        r.printArea();
        Triangle t = new Triangle();
        t.x = x1;
        t.y = y1;
        t.printArea();
        Circle c = new Circle();
        c.x = x1;
        c.printArea();
        sc.close();
    }
}
```

OUTPUT:

Enter Length and Breadth: 10

5

Area of Rectangle is 50.0

Area of Triangle is 25.0

Area of Circle is 314.2857